Artificial Intelligence for Digital Characters

Speech Synthesis II

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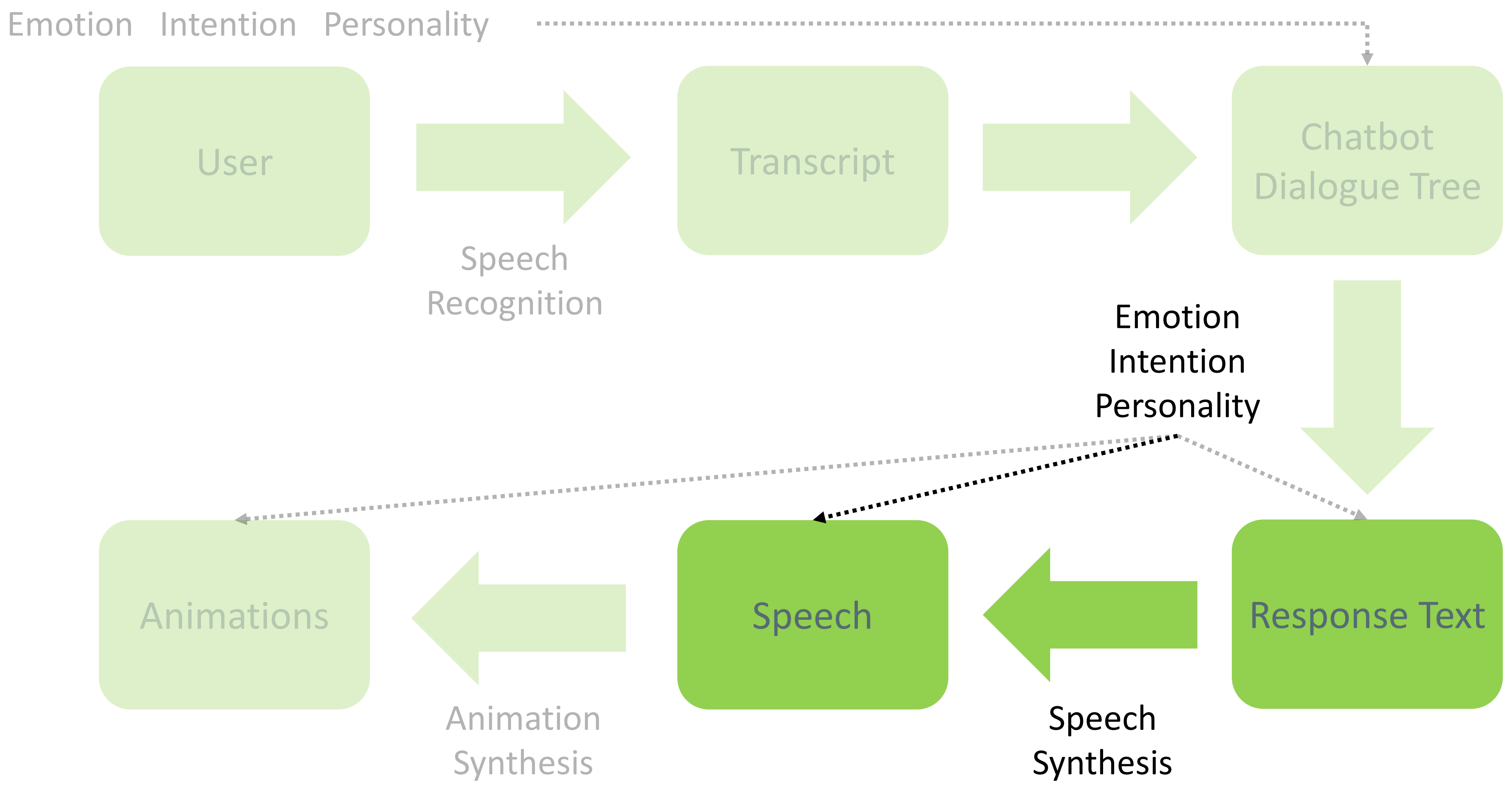
Conversational Digital Character



Speech Synthesis

Life is like riding a bicycle. To keep your balance, you must keep moving.

Text Analysis

Linguistic Features

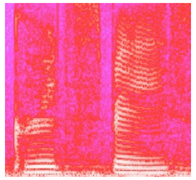
Acoustic Model

Acoustic Features

Vocoder

Waveform

Text Normalization
+
Phonemization



Acoustic Model

Vocoder

Style Transfer

Evaluation of TTS

Tacotron 2

FastSpeech 1/2

Uni-TTS

Speech Synthesis

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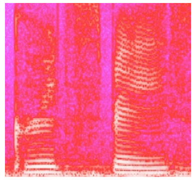
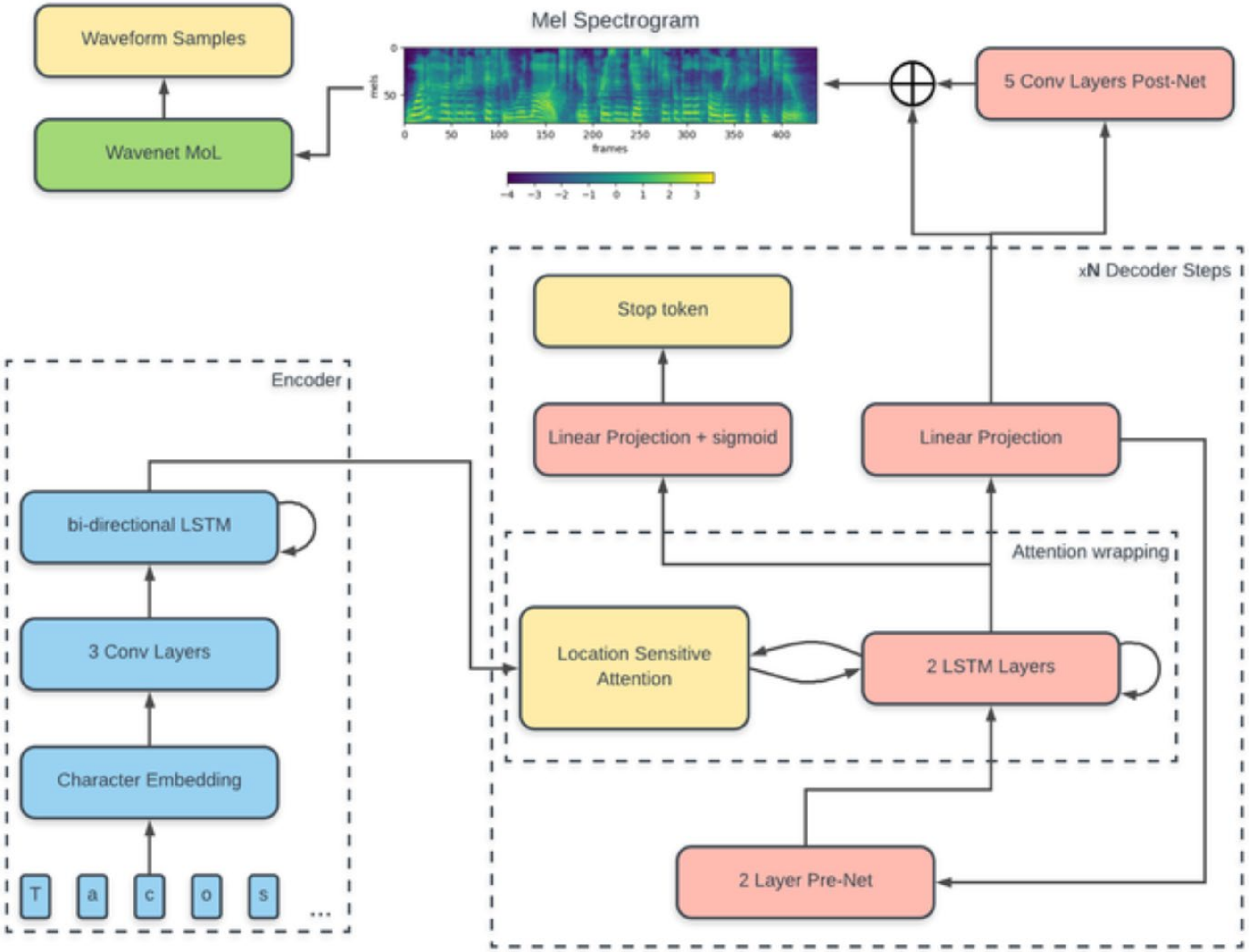
Text Analysis → Linguistic Features → Acoustic Model → Acoustic Features → Vocoder → Waveform

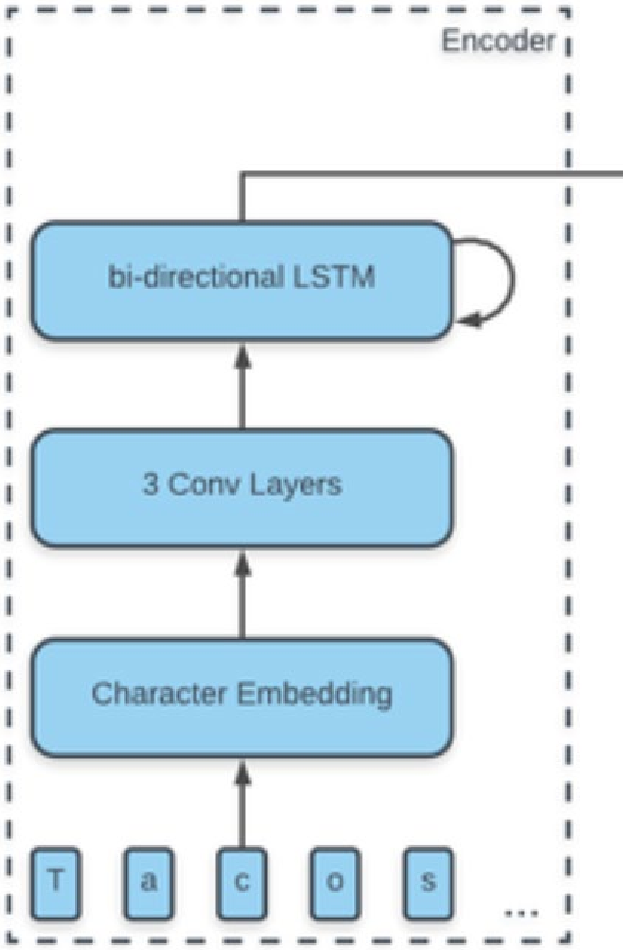


Tacotron 2



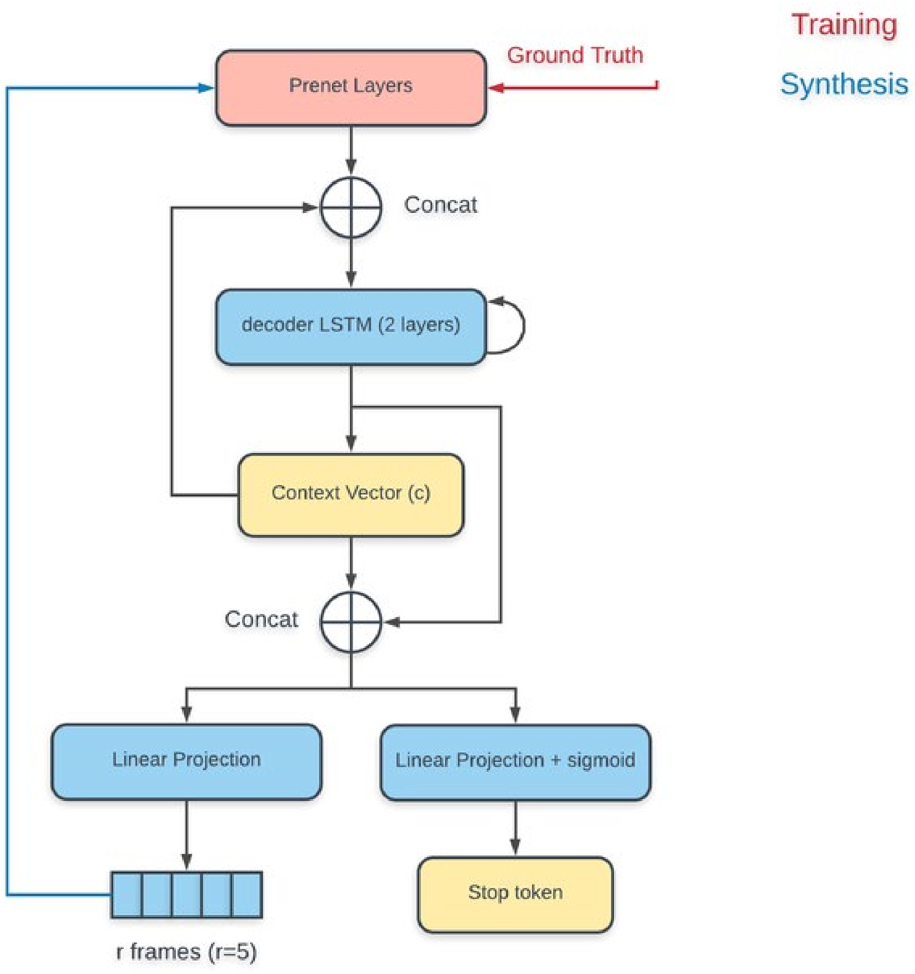
[Shen et al. 2018]

Tacotron 2: Encoder



- Input (characters): $X = \{x_1, x_2, \dots, x_t\}, X_i \in \mathbb{R}^k$
- Output (hidden states): $H = \{h_1, h_2, \dots, h_t\} \in \mathbb{R}^l$
- Character Embedding
  - 512 dimensional vector per character
  - Randomly initialized
- Convolution Layers
  - 512 filters with length of 5
  - Captures two previous and next characters
- Bi-directional LSTM
  - Info about past and future
  - Concat forward and backward hidden state

Tacotron 2: Decoder



- Prenet
  - Two fully connected layers (256 neurons)
- Postnet
  - Input: y_i, Output: $y_{final,i}$
  - Convolutional layers followed by FNN (80 neurons)
  - Improves output quality
- Reduction factor r
  - Decoder predicts r frames
  - Speeds up computation and reduces memory usage
- Loss

$$L = \frac{1}{n}\sum_{i=1}^{n}(y_{real,i} - y_i)^2 + \frac{1}{n}\sum_{i=1}^{n}(y_{real,i} - y_{final,i})^2$$

Tacotron 2: Examples

Complex words

Generative adversarial network or variational auto-encoder.

Phrase semantics

He has read the whole thing.

He thought it was time to present the present.

Spelling errors

Thisss isrealy awhsome.

Punctuation

This is your personal assistant, Google Home.

This is your personal assistant Google Home.

Intonation

The buses aren't the PROBLEM, they actually provide a SOLUTION

Questions

Does the quick brown fox jump over the lazy dog?

https://google.github.io/tacotron/publications/tacotron2/

Tacotron 2 or Human?

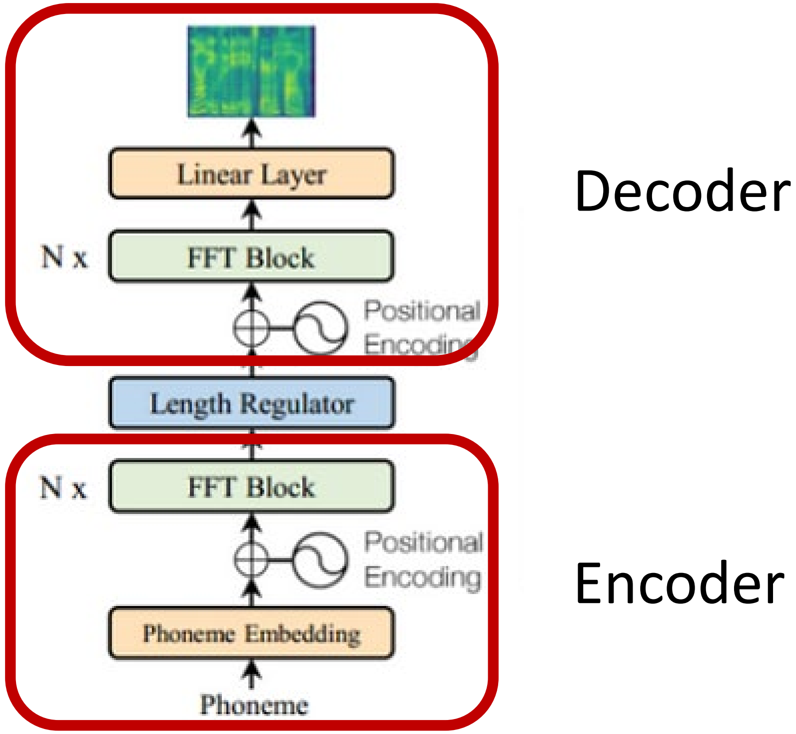
Human | Tacotron 2 | Tacotron 2 | Human

https://google.github.io/tacotron/publications/tacotron2/

Tacotron 2: Drawbacks

- Slow inference speed
- Prone to word skipping and repetition
- Lack of control features (speed, prosody)

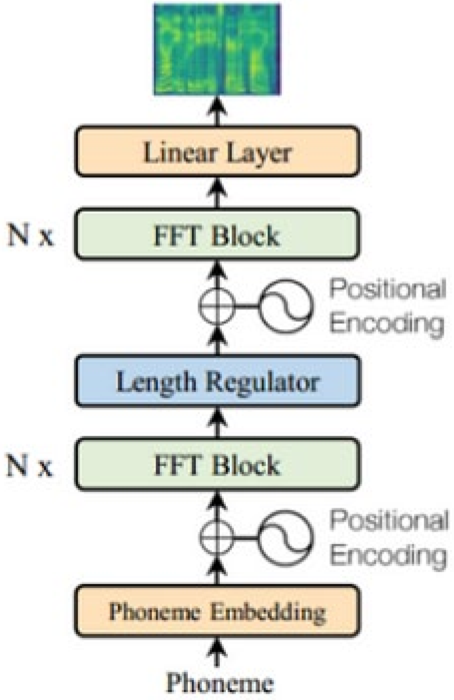
FastSpeech: Explicit Duration Modelling



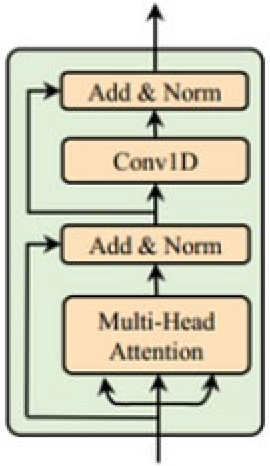
- Text-to-phoneme conversion (last lecture)
- Encoder
  - Input: $[x_1, \ldots, x_n], x_i \in \mathbb{R}^d$
  - Output: $[h_1, \ldots, h_n], h_i \in \mathbb{R}^e$
- Decoder
  - Input: $[h_1, \ldots, h_p], h_i \in \mathbb{R}^e$
  - Output: $[y_1, \ldots, y_p], y_i \in \mathbb{R}^{80}$
- Phoneme Embedding
  - High dimensional vectors $x_i \in \mathbb{R}^d$
  - Learned during training
- Feedforward Transfomer (FFT)
  - 6 blocks

Ren et al 2019

FastSpeech: Explicit Duration Modelling



FFT Block

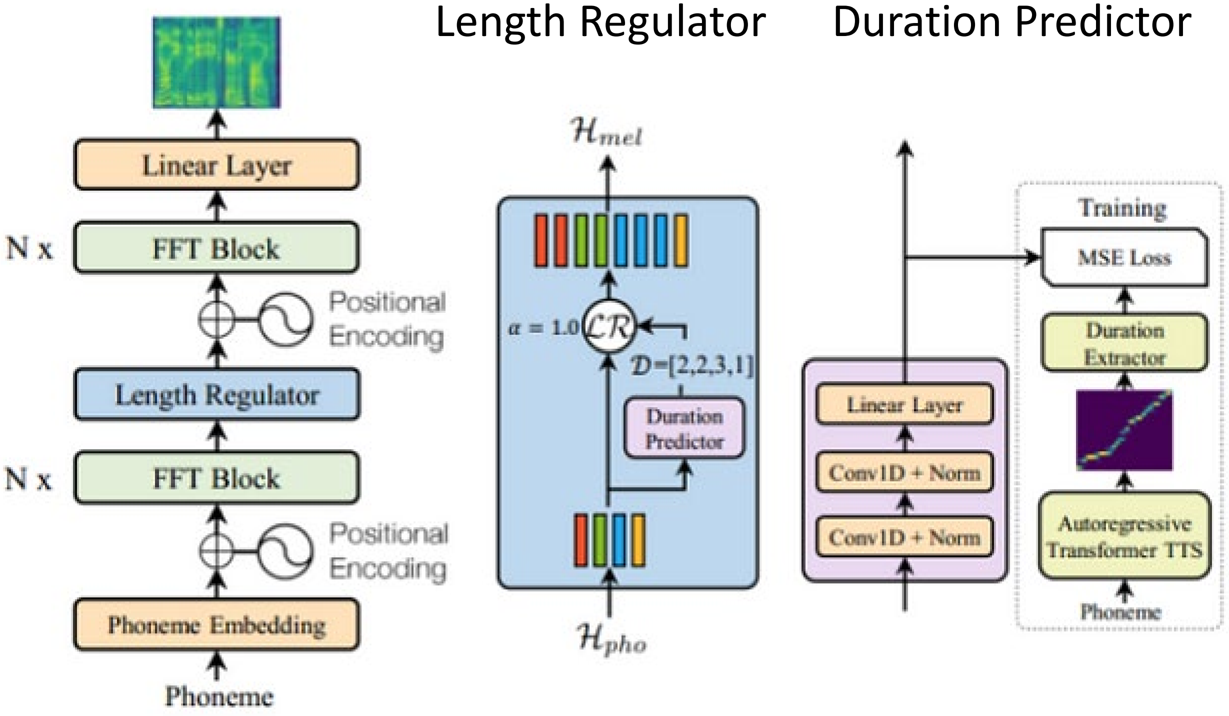


- Self-attention
- 1D convolution (instead of FFN)

Ren et al 2019

FastSpeech: Explicit Duration Modelling

Length Regulator Duration Predictor



- One phoneme corresponds to several mel-spectrograms columns
- Upsample by repeating encoder states by the predicted duration
- Example
  - Word *knight*
  - D = [1, 8, 15, 3, 0, 17]
- Control speed by scaling duration
  - 2*D = 0.5x speed up
  - 0.5*D = 2x speed up
- Duration predictor trained based on duration predictions from other TTS

Ren et al 2019

FastSpeech: Examples

Speed 1.00x

Speed 1.25x

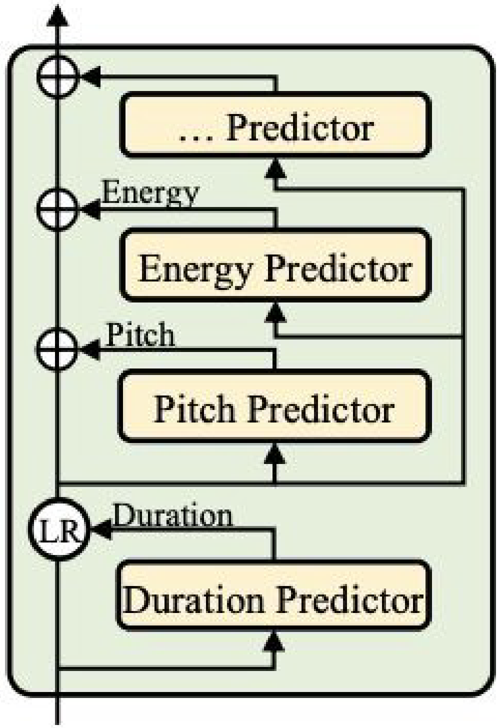
Speed 0.75x

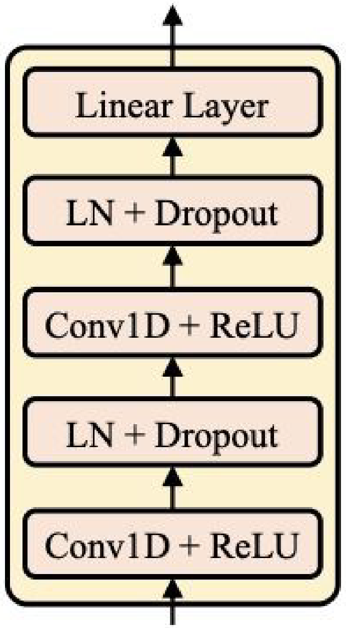
No breaks

Breaks

https://speechresearch.github.io/fastspeech/

FastSpeech 2 Variance Predictors

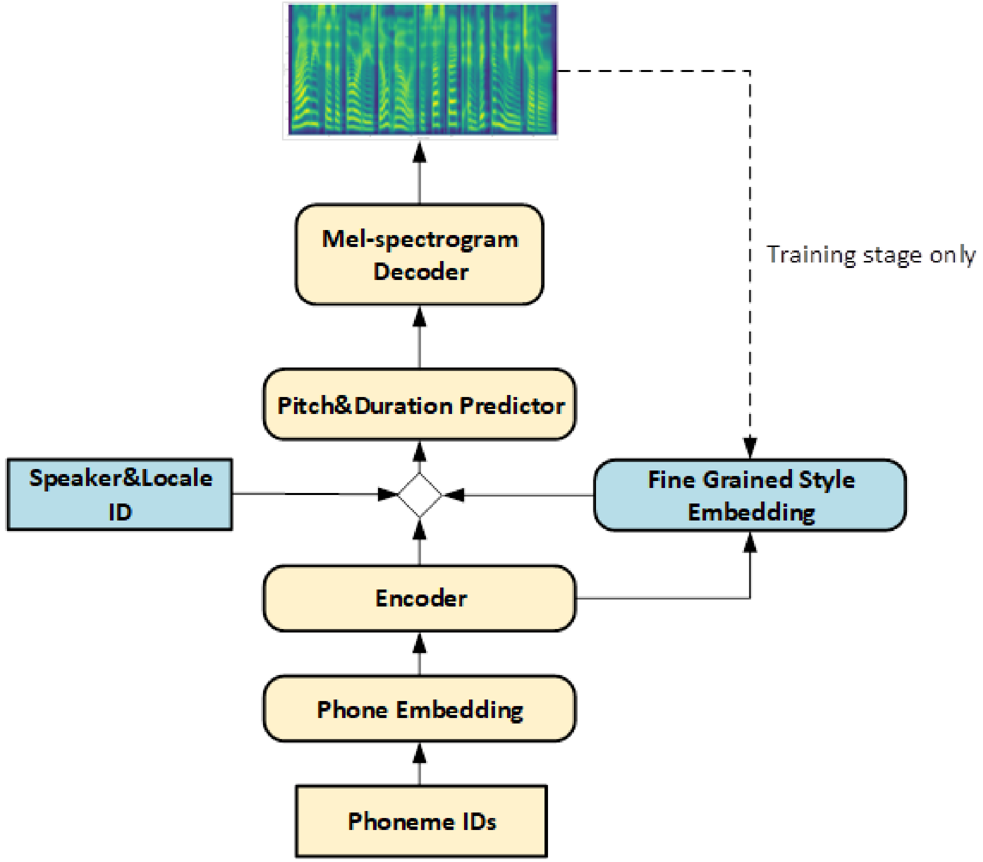




At training time use the ground truth duration, energy, and pitch for synthesis and train predictors with MSE

[Ren et al. 2020](#)

Uni-TTS (Microsoft Azure TTS)



- Based on FastSpeech 2
- Base model trained on 3,000 hours of recordings (multiple speakers, multiple languages)
  - Can be fine-tuned for target speaker
- Style embedding
  - Boost naturaleness of speech
  - Fine-grained prosody from spectrogram (training)
  - Predicted by encoder (inference)
- Speaker ID & Local ID
  - Controlling timbre and accent

Kang et al. 2021

Tips For Training TTS Model

- Trim samples, normalize volume, remove noisy samples
- Normalize Log Mel Spectrograms
  - Min-Max Normalization
  - Standardization
- Fine-tune from existing models
  - Can be useful for small/noisy datasets

Acoustic Model

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Evaluation of TTS

Griffin-Lim

WaveNet

HifiGAN

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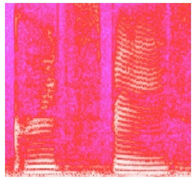
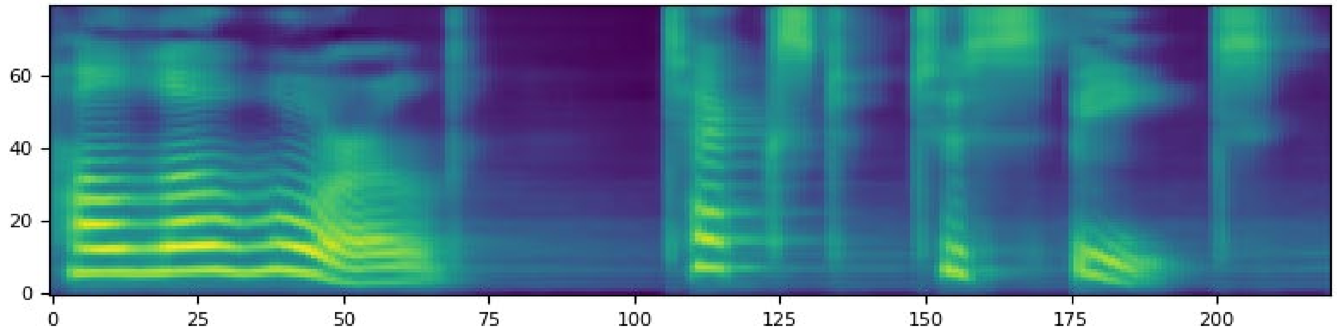


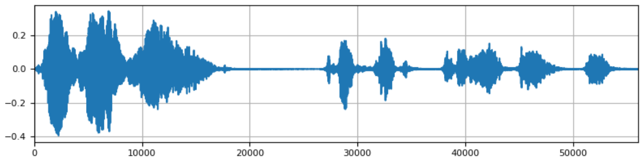
Vocoder

Waveform

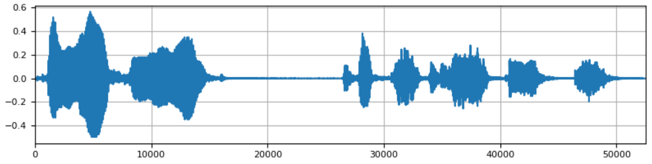
Different Vocoders



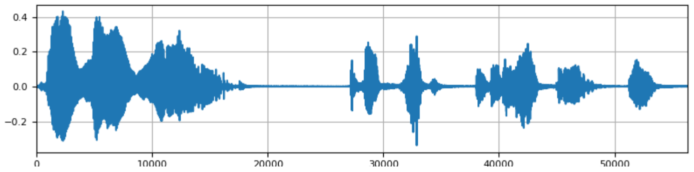
Griffin-Lim



WaveNet



HiFiGAN



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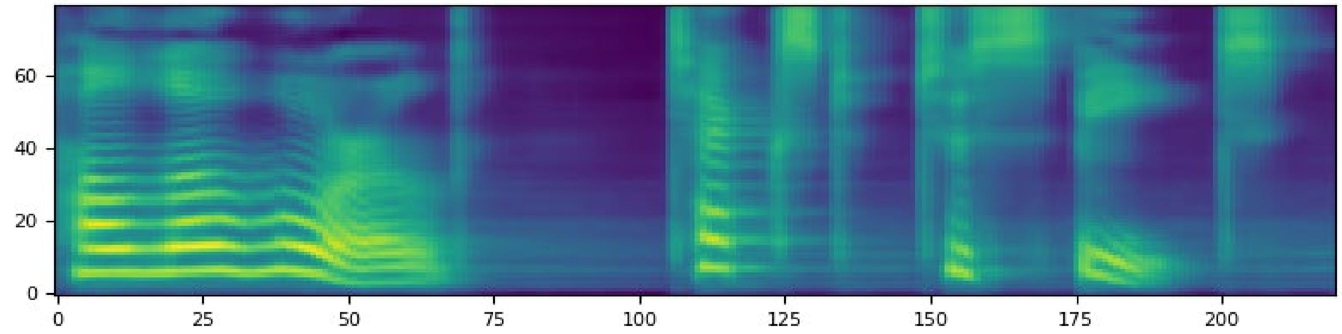
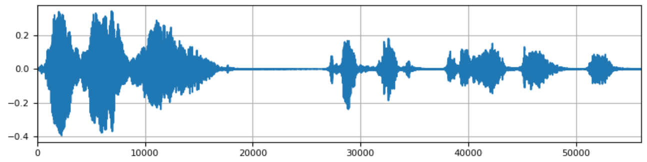


Griffin-Lim



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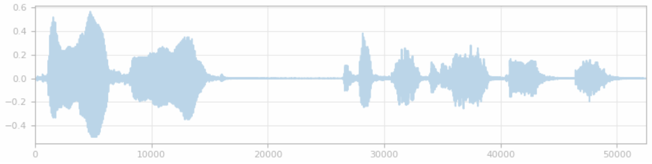
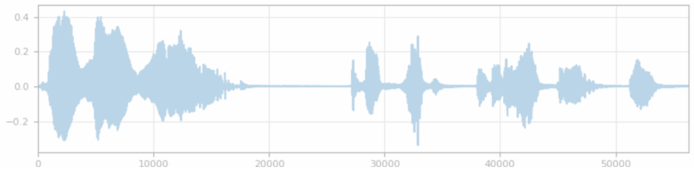


HiFiGAN



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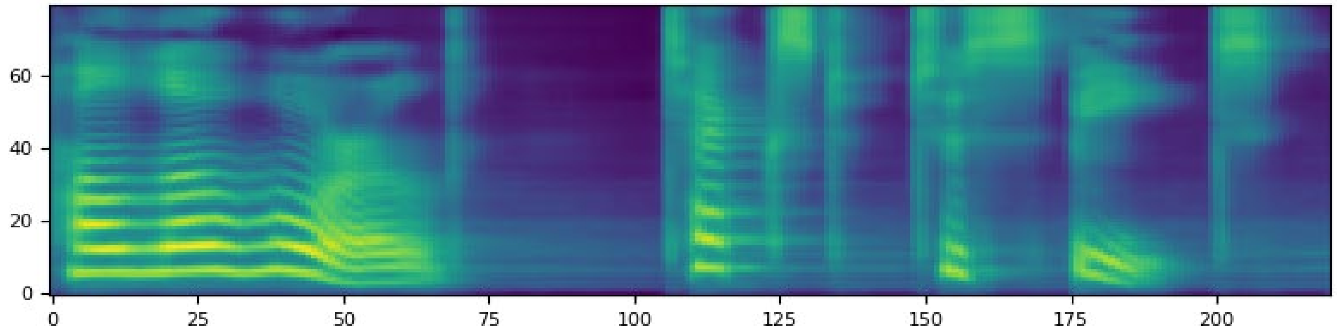

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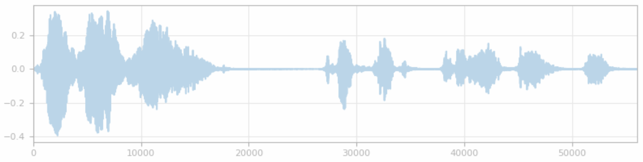

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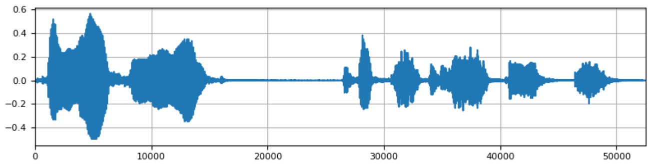

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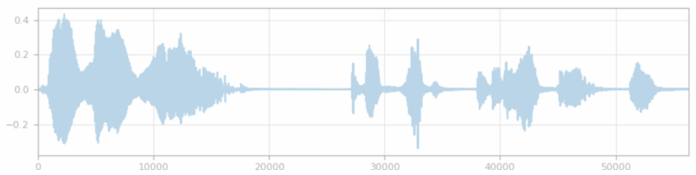

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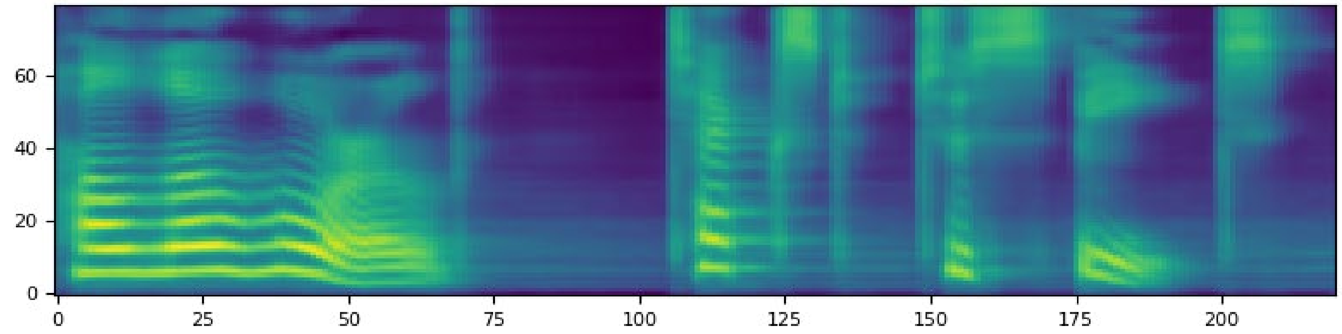

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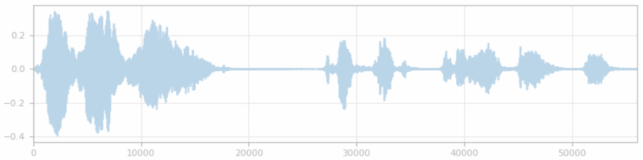

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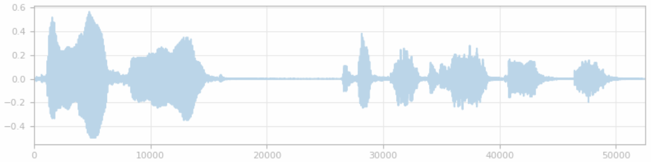

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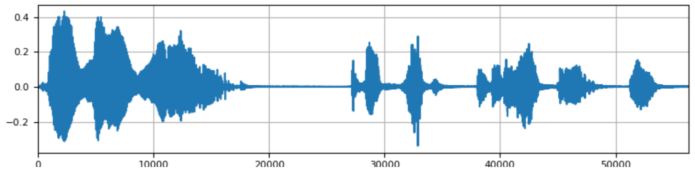


Phase Prediction

- We have a magnitude log mel spectrogram
- To construct waveform, we need complex spectrogram
  - Magnitude
  - Phase
- Apply inverse STFT (iSTFT)
  - Reconstruct time-domain signal
  - Overlapping windowed inverse Fourier transforms of the spectrogram frames
  - Adding overlapping segments of reconstructed signal together

Griffin-Lim

- Iteratively reconstructs phase information from just the magnitude spectrogram
  - Pure signal processing
  - No learned parameters
- Algorithm
  1. Start with a random prediction for the phase
  2. Convert the complex spectrogram into the time-domain signal using iSTFT
  3. Apply STFT on the time-domain signal
  4. Replace the magnitude by the initial magnitude
  5. Repeat steps 2 to 4 until difference in phase is below threshold

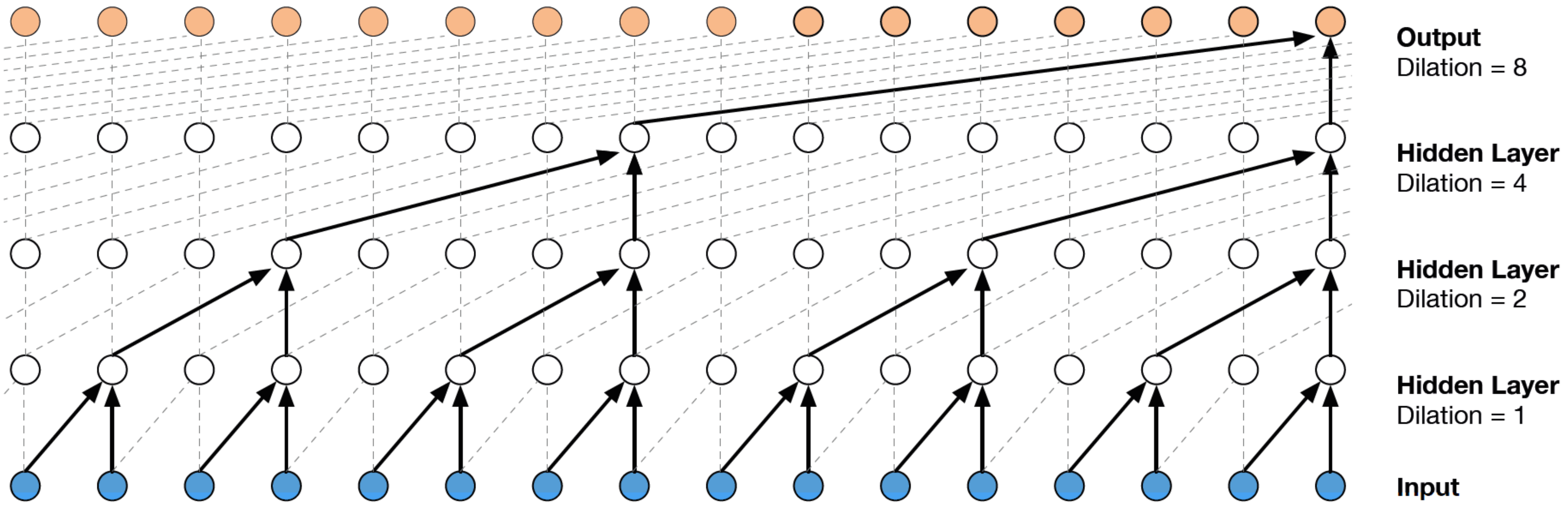
Griffin, Lim 1983

WaveNet

- Auoregressive: generates one sample of audio at a time
- Very high output quality but very slow
- Key components
  - Causal dilated convolutions: capture long-term dependencies
  - Softmax at output: classification rather than regression (8-bit mu-law)

Oord et al. 2016

WaveNet: Dilated Convolutions



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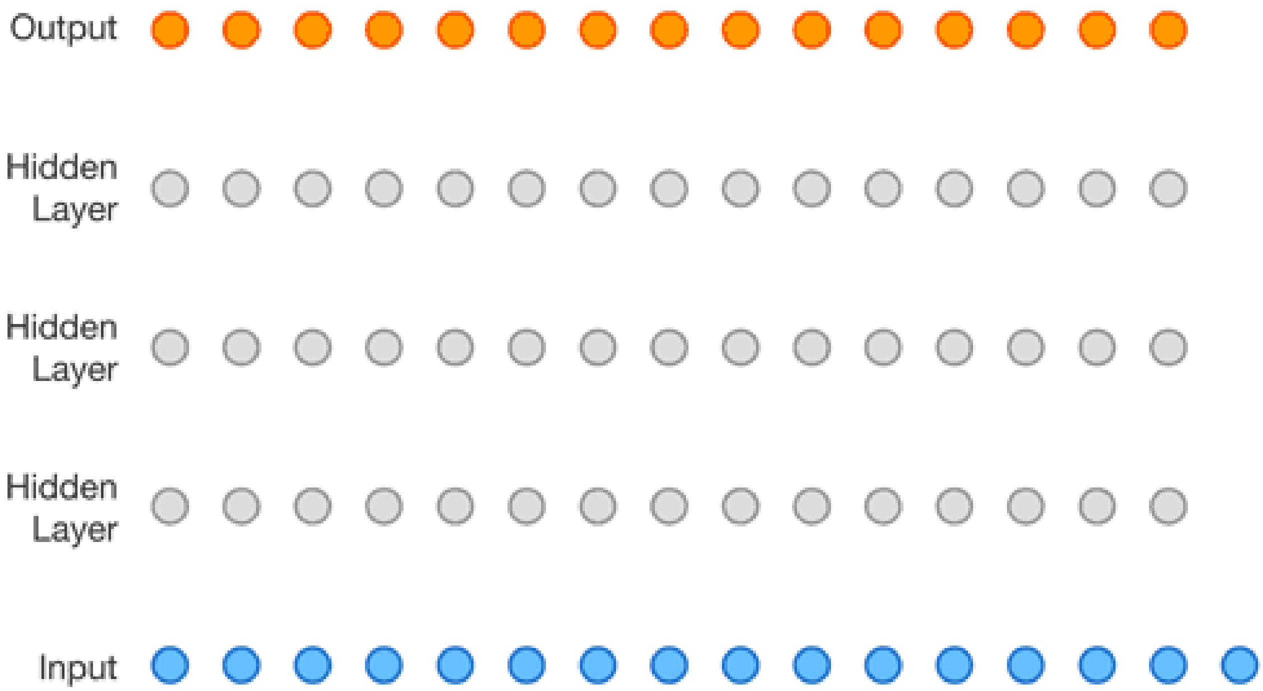


Other Vocoders

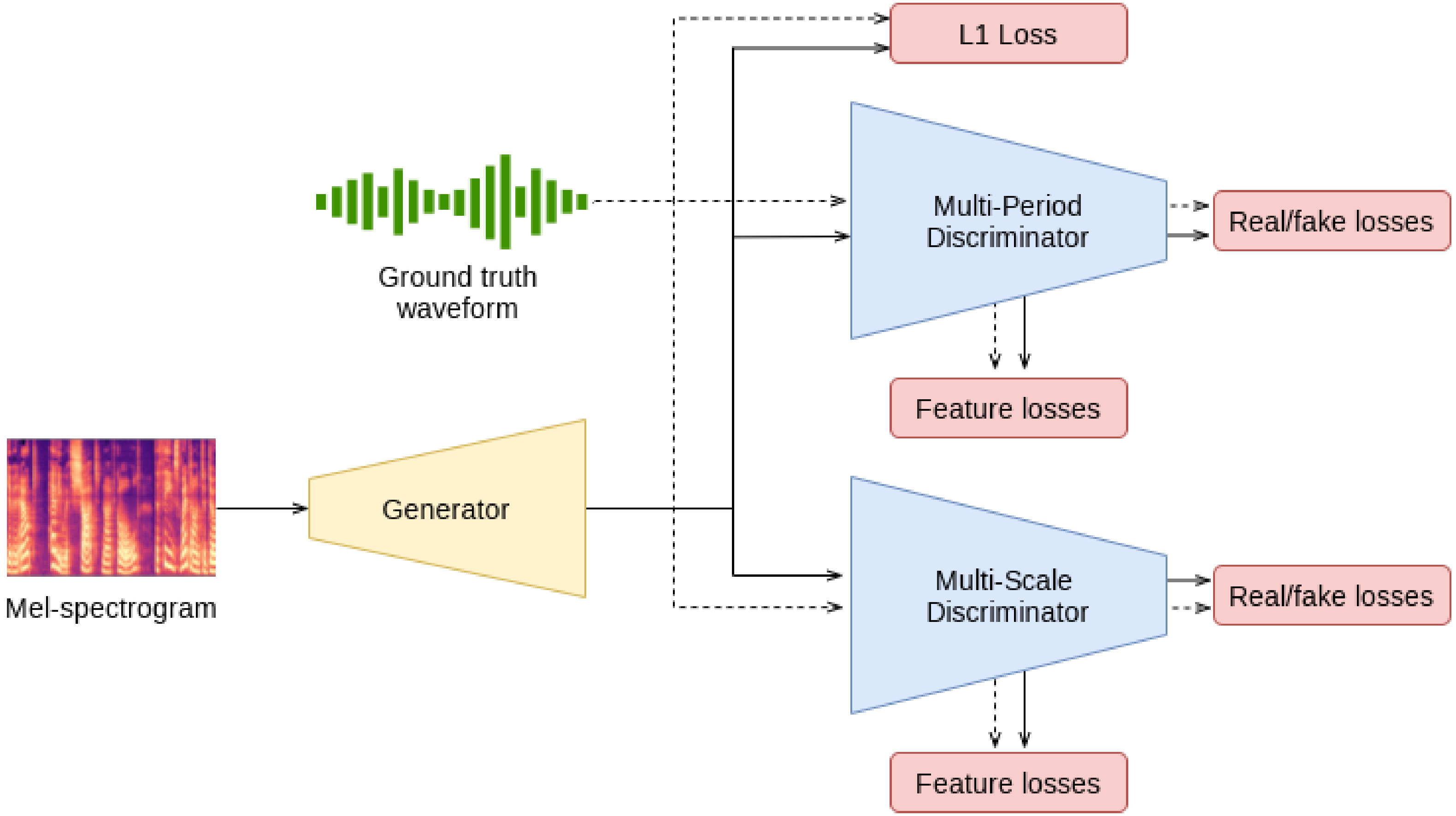
- WaveRNN (DeepMind) Kalchbrenner et al. 2018
  - Simple, autoregressive GRU
  - Runs on smartphone CPUs
  - Requires heavy optimization
- WaveGlow (NVIDIA) Prenger et al. 2018
  - Flow-based
  - Generates high-quality speech
  - Simple and fast
  - Requires powerful GPU for fast inference

HiFiGAN

- Similar vocoder used by Microsoft Azure TTS
- Excellent sound quality at low computational cost
- Simulatenously train two networks: a generator and a discriminator
- Generator produces audio from the spectrograms to be as close as possible to the ground truth audio
- Discriminator trained to distinguish generator outputs from real audio
- Other similar models
  - MelGAN, Parallel WaveGAN

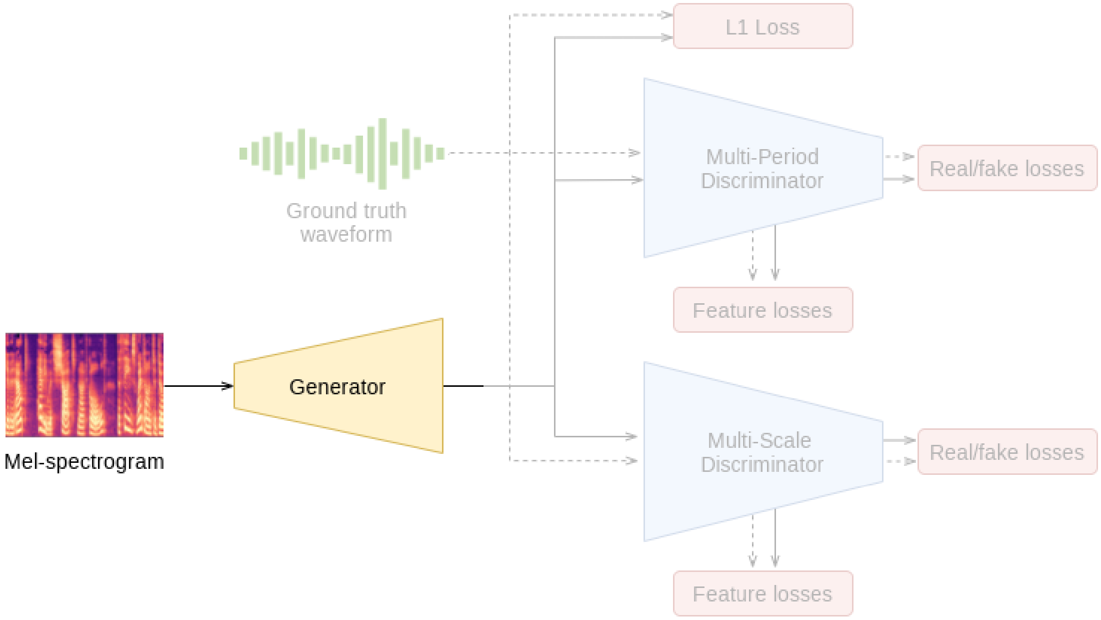
Kong et al. 2020

HiFiGAN



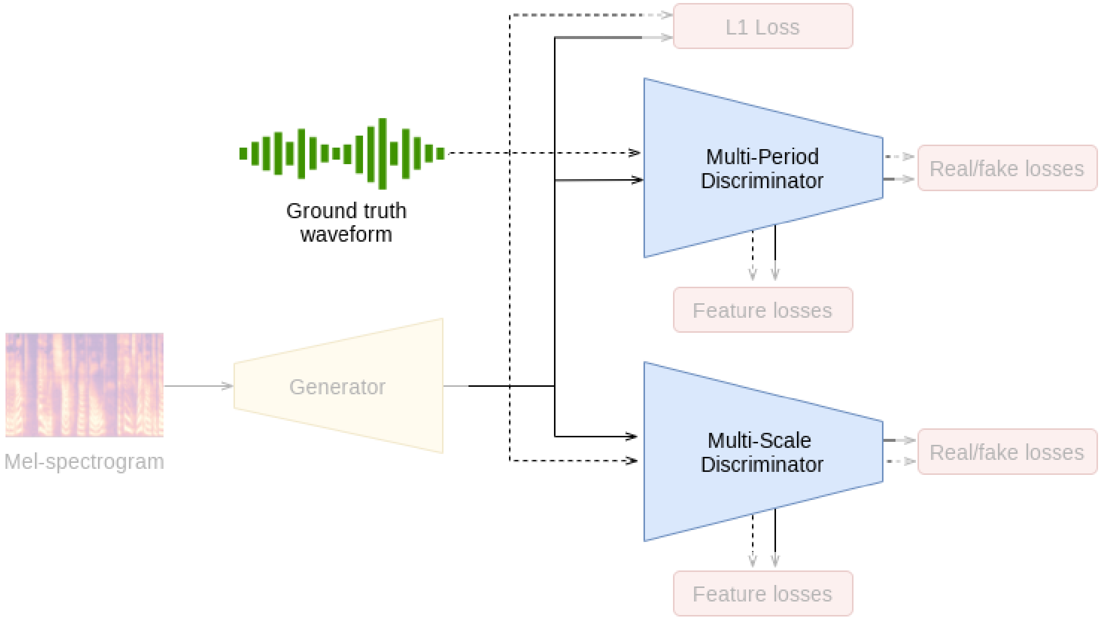
L1 Loss
Multi-Period Discriminator
Real/fake losses
Ground truth waveform
Feature losses
Mel-spectrogram
Generator
Multi-Scale Discriminator
Real/fake losses
Feature losses

HiFiGAN



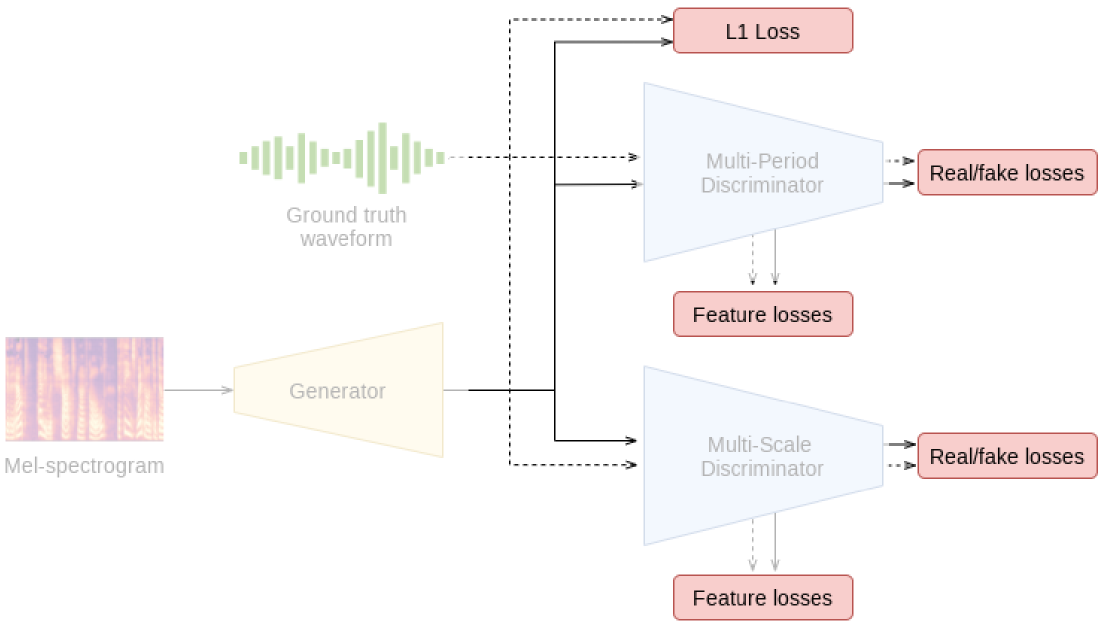
- Upsampling layers
  - Increasing temporal resolution
  - Transposed convolutions
- Residual blocks
  - Dilated convolutions
  - Nonlinear activation functions
  - Residual connections
- Multi-receptive Field Fusion
  - Parallel residual blocks with different dilation rates
  - Outputs of residual blocks fused

HiFiGAN



- Multi-Scale Discriminator
  - Several discriminators (2-3)
  - Capture wide range of details
  - Downsample signal
- Multi-Period Discriminator
  - Analyzes periodic nature of audio
  - More accurate pitch and tone
  - E.g. Fourier Transform

HiFiGAN



- Real/fake loss
  - Binary cross-entropy loss
  - Discriminator: Correctly identify real and fake samples
  - Generator: Success in tricking discriminator
- Feature loss
  - Distance between features (L1 or L2)
  - Features from intermediate layers
  - Encourages realistic audio
- L1 loss (Mel-Spectrogram Loss)
  - Difference of mel-spectrograms of generated audio and real audio
  - Ensures spectral characteristics match

Acoustic Model

Vocoder

Style Transfer

Evaluation of TTS

One Hot Labels

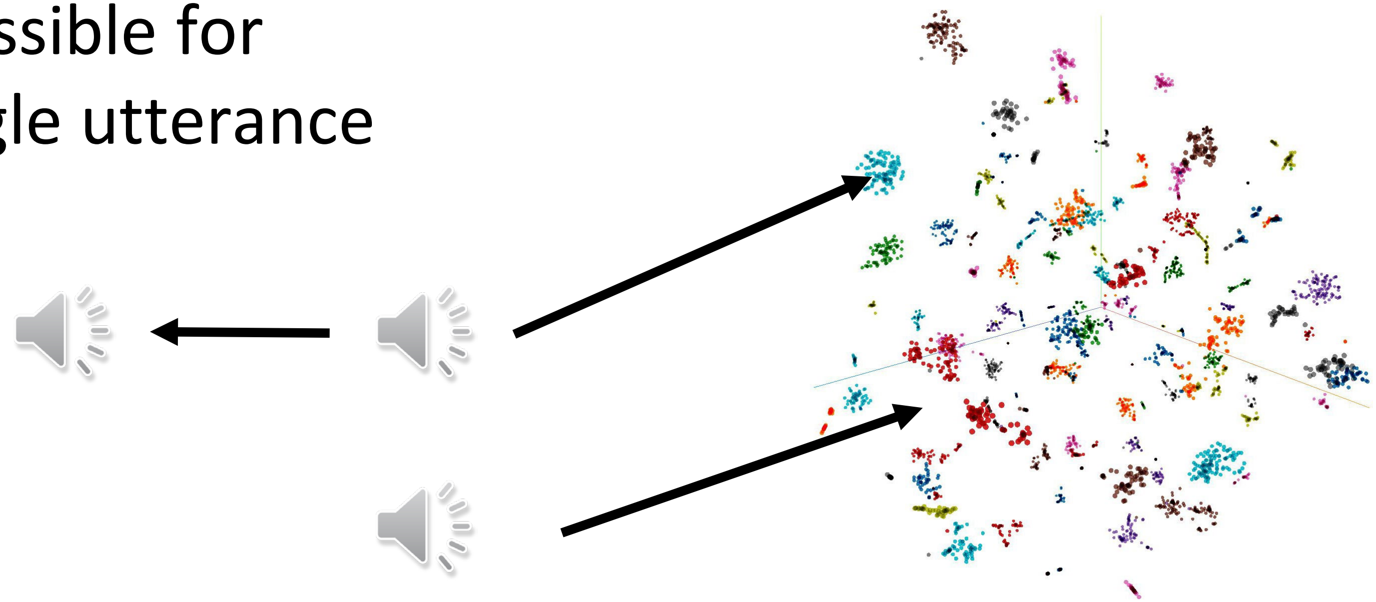
Learned Embeddings

Speaker/Style with One Hot Labels

- Enumerate speakers and/or styles and label training data
- Training: learn an embedding for each speaker/style by passing a one hot encoding to the encoder
  - Example: [0, 0, 1, 0, 0] for speaker 3
- Inference: Pass in the corresponding speaker/style embedding
- Simple and easy to train but constrained by the breadth of labels

Learned Speaker Embeddings

- Learn speaker embedding with large datasets of speaker-labelled audio
  - CNN or RNN
- Feed frozen embeddings to TTS model at training and inference time
- If the training dataset is sufficiently diverse, zero shot synthesis is possible for new speakers with a single utterance



https://google.github.io/tacotron/publications/speaker_adaptation/index.html

Learned Style Embeddings

- Instead of explicitly labelling style, can we learn the style in the audio data automatically?
  - Reduces labor-intensive labelling
  - Discovering subtle or complex styles (not possible manually)
- Training: Feed in the mel spectrogram to style module
  - CNN, LSTM
- Inference: Feed in a reference mel spectrogram or sample from latent space

Style Examples (Microsoft Azure TTS)

Cheerful

Sad

Angry

Excited

Friendly

Unfriendly

Hopeful

Terrified

Shouting

Whispering

Different Style Degrees

Style degree 1.0

Style degree 2.0

Acoustic Model

Vocoder

Style Transfer

Evaluation of TTS

Intelligibility Test

Diagnostic Rhyme Test

Modified Rhyme Test

Recap: Word Error Rate

How to evaluate the word string output by a speech recognizer?

$$\text{Word Error Rate} = \frac{100 * (\text{Insertions} + \text{Substitutions} + \text{Deletions})}{\text{Total Word in Correct Transcript}}$$

REF: portable form of stores last night so
SYS: portable phone upstairs last night so

Evaluation of TTS

- Evaluation of TTS generally requires humans!
  - Listen to example utterances, rate various aspects (naturalness, intelligibility, friendliness, expressiveness, etc.). Scale of 1-5
  - Mean opinion score (MOS). Average of ratings
  - AB Tests (prefer A, prefer B)
- Intelligibility Tests
  - Did the human hear the correct thing?

Evaluation of TTS

- Diagnostic Rhyme Test (DRT)
  - Humans listen identification choice between two words differing by a single phonetic feature
  - 96 rhyming pairs
  - Veal/feel, meat/beat, vee/bee, zee/thee, etc
    - Subject hears "veal", chooses either "veal or "feel"
    - Subject also hears "feel", chooses either "veal" or "feel"
  - Often embedded in carrier phrases (now we will say <word> again)
  - % of right answers is intelligibility score
- Modified Rhyme Test (MRT)
  - 50 sets of 6 words
  - Example set: went, sent, bent, dent, tent, rent
  - Listeners hear single word and must choose one word in set
- Reading addresses out loud, reading lines of news text, etc.

Summary

Acoustic Model
- Generates Mel-spectrograms from text
- Attention-based models (Tacotron 2) & Duration-based models (FastSpeech, Unit-TTS)
- Preprocessing training data important

Vocoder
- Generates waveform from Mel-spectrogram
- Different architectures (Griffin-Lim, WaveNet, HiFiGAN, DiffWave)
- HiFiGAN best trade-off between quality and speed

Style Transfer
- One hot encodings simple but need labels
- Learning embeddings automatically from data

Evaluation of Chatbots
- Qualitative evaluation
- Quantitative evaluation

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